

Java Looping Problems

For Loop Problems:

1. Write a program to print the first 10 natural numbers using a for loop.
2. Create a program that calculates the factorial of a number using a for loop.
3. Develop a program to print the sum of the first 100 even numbers using a for loop.
4. Write a program that prints a multiplication table for a given number using a for loop.
5. Write a program to display numbers from 1 to 50 but skip numbers that are multiples of 3 using a for loop.

While Loop Problems:

1. Create a program that finds the sum of all odd numbers up to 100 using a while loop.
2. Write a program to reverse a given number using a while loop.
3. Develop a program to count the number of digits in a given integer using a while loop.
4. Write a program to display Fibonacci sequence up to a given number using a while loop.
5. Create a program that keeps taking input from the user until a negative number is entered.

Do-While Loop Problems:

1. Write a program that repeatedly asks the user for input until they enter 'exit', using a do-while loop.
2. Create a program to print the sum of all integers entered by the user until a zero is entered using a do-while loop.
3. Develop a program that displays the first 20 terms of a geometric sequence using a do-while loop.
4. Write a program to print a countdown from a given number to 1 using a do-while loop.
5. Create a program to calculate the average of numbers entered by the user until they enter -1, using a do-while loop.

Mixed Loop Problems Problems:

1. Write a program to determine if a number is prime using any loop.
2. Create a program that prints all prime numbers between 1 and 100 using any loop.
3. Write a program to generate Pascal's triangle up to a given number of rows using nested loops.
4. Develop a program that prints a diamond shape pattern with asterisks using loops.
5. Create a program to find the greatest common divisor (GCD) of two numbers using a while loop.